Assignment 1

Group 24

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1 Part 1: Vertical variation of wind speed

1.1 Met Mast and Location

Task: Describe the met mast and its location. How do the surroundings of the mast look in terms of terrain roughness? How far is the mast from the turbine and in which direction? Why?
The measurements are from DTU Risø Campus using a met mast with a total height of 70 m. The mast is around 120 m, away from the Vestas V52 wind turbine. Regarding the terrain, the location is characterized by varying roughness lengths. To the west, it's dominated by the Roskilde Fjord, which leads to very low terrain roughness. In the other directions, the roughness is higher due to a mix of open fields with light vegetation and the campus buildings. The mast is positioned at this distance and orientation because the prevailing winds in the area come from the west, over the fjord. This setup allows for the measurement of undisturbed wind conditions, as the mast is located upwind of the turbine, ensuring that the data collected is not influenced by the wake effects of the turbine. Therefore free flow measurements can be obtained.

1.2 Data Collection

Task: Describe the data collected. What has been measured, what units, what sample rate, where at the mast and with what instruments?
The data collected on February 10th, 2026, corresponds to three categories: **Wsp, AirAbs, Sdir**. They correspond to horizontal wind speeds measured with cup anemometers and wind vanes in m/s^2, wtf is AirAbs and what is its unit?, and southern wind direction in .
Since the wind is a dynamic fluid, each category varies according to height prompting them to be measured at different rungs of the mast. Wsp is measured at 70 m (i.e. the uppermost level of the mast), 57 m, 44 m, 31 m, and 18 m; then, AirAbs is measured at 70 m and 18 m; and lastly, Sdir is measured at 70 m, 44 m, and 18 m.
All measurements, i.e. the three categories at their respective mast heights, are simultaneously recorded with 10-minutes resolution.

1.3 Raw Time Series Plot

Task: Plot the raw time series of the wind speed at the different heights. What time scales do you see? Do they differ for the different heights?

Figure 1: Raw wind speed time series at various heights (10th Feb 2026).

1.4 Logarithmic Wind Profile

Task: Plot the average wind speed for each height against height z. Can the profile be described by a logarithmic wind profile?

1.5 Roughness Length and Friction Velocity

Task: Estimate the roughness length z_0 and the friction velocity u_* from the observed wind profile.

1.6 Stability Impact

Task: Discuss the validity of the logarithmic profile under the observed stability conditions.

1.7 Heat Flux and Stability

Task: Estimate the friction velocity and the heat flux from the sonic anemometer data.

1.8 Weather Conditions

Task: How was the weather at the day of visit? Does the observed weather support your finding of question 7?
The day we visited Risø was windy from the eastern direction with some snowflakes. Generally, the site experiences western winds from the fjord, but that was not the case that day.
does it support finding of Q7?

2 Part 2: Statistical analysis of multi-year wind climate

2.1 Data Description

Task: Describe the data. What has been measured, in what units, at which locations, with which instruments and with what temporal resolution?
The dataset consists of eight years of wind observations from 2018 to 2024, provided as 10-minute mean values. The data in the file `tenMinutesData_2018_2024.csv` originates from the meteorological mast at the DTU Risø Campus.

Parameter	**Symbol**	**Unit**	**Instruments / Label**
Wind speed	sp	[m/s]	Cup (W) and Sonic (SW) anemometers
Wind direction	dir	[°]	Wind vanes (W) and Sonic (SW) sensors
Precipitation	Rain	[mm]	Tipping bucket rain gauge

Table 1: Summary of measured parameters and sensor types.

The measurements are available at specific heights, including 57 m and 70 m for wind speed, and 41 m for wind direction. For the long-term climate analysis and the Annual Energy Production (AEP) calculation of the V52 turbine, the data collected at the 70 m height is of primary interest.

2.2 Raw Data and Gaps

Task: Plot the raw data as function of time. Identify any data gaps and discuss possible reasons for these gaps.

Figure 2: Long-term wind speed time series (2018-2024) highlighting data gaps.

2.3 Seasonal Wind Roses

Task: Using the whole time series, plot a wind rose for the winter months (Dec-Feb) and one for the summer months (Jun-Aug) discuss the directional distribution of the wind. Does it match what you would expect for Denmark?

2.4 Histogram and Weibull Fit

Task: Define wind bins with a spacing of 1m/s and plot a histogram of the wind speed variation based on the data from 70m height for two different years, 2018 and 2024. Fit a Weibull distribution to the observed wind speed.

$$f(u) = \frac{k}{A}\left(\frac{u}{A}\right)^{k-1} \exp\left[-\left(\frac{u}{A}\right)^{k}\right] \tag{1}$$

2.5 AEP for V52 Turbine

Task: Calculate the AEP for the V52 turbine located at the met mast site, based on the 2018 and 2024 wind measured data. Discuss implications on the yearly variability.

Year	AEP [MWh]	Capacity Factor [%]
2018	...	...
2024	...	...

Table 2: Annual Energy Production (AEP) and Capacity Factor for the V52 turbine.